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RESEARCH

Effect of class IV LASER therapy along with Low-dye kinesio-taping technique or plantar fascia stretching on pain, pain pressure threshold, and disability in individuals with plantar fasciitis: Double-blinded randomized clinical trial

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Abstract

The purpose of this study was to determine which adjunctive intervention, low-dye kinesio-taping or plantar fascia stretching, when combined with Class IV LASER therapy, provides greater improvement in pain intensity, pain pressure threshold (PPT), and functional disability among individuals with plantar fasciitis (PF). In this double-blinded randomized clinical trial, 50 subjects (mean age 29.72 ± 7.47; 13 males and 37 females) with PF, were randomly assigned into an experimental group (class IV LASER therapy + low-dye kinesio-taping technique) and a control group (class IV LASER therapy + plantar fascia stretching) using computer-generated block randomization without sex stratification. Blinding was ensured for both the participants and the outcome assessor. Treatment was administered 3 times a week for 2 weeks. Outcome measures included the Visual Analogue Scale (VAS) measuring pain intensity, Pressure Algometry for PPT, and Foot Function Index (FFI) for disability. Assessment was done at baseline and after 2 weeks of intervention. Normality was tested with the Kolmogorov–Smirnov test, and statistical analysis included paired t-tests for change within groups and independent t-tests for group differences. Effect sizes were large ($d = 1.25$ for VAS, 2.48 for PPT, and 0.69 for FFI), corresponding to 62.3%, 48.6%, and 54.2% improvements, respectively, all exceeding established minimally clinically important difference (MCID) thresholds. Significant improvement was observed in VAS, PPT, and FFI scores within both groups (p-value < 0.05), and between-group analysis showed a significant difference (p-value < 0.05) in all outcome measures, in favour of the control group. Cohen's d was observed to be between 1.4 and 2.2 for primary outcomes. Class IV LASER therapy, when combined with plantar fascia stretching, demonstrated superior outcomes compared to its combination with low-dye kinesio-taping, resulting in greater pain reduction, improved PPT, and enhanced foot function in individuals with PF. CTRI/2023/05/053275.

Keywords Fascia · Humans · LASER therapy · Plantar fasciitis · Pain

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Abbreviations

VSL	Visual Analogue Scale
FFI	Foot Function Index
PPT	Pain Pressure Threshold
ESWT	Extracorporeal Shockwave Therapy
LASER	Light Amplification by Stimulated Emission of Radiation

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Introduction

Plantar fasciitis, also called plantar fasciopathy, is a common condition affecting the foot, characterized by symptoms such as heel discomfort, sensitivity, and impaired foot function [1]. It is estimated that approximately 10% of individuals will experience heel pain related to this condition during their lifetime. Moreover, roughly 10–16% of individuals may suffer from heel pain at least once, with 20–30% of them experiencing it in both feet. Notably, the highest incidence of plantar fasciitis is seen among adults aged 40 to 60, regardless of gender [2, 3].

This condition can significantly hinder a person's daily activities due to increased pain and tenderness, affecting foot function [4]. The initiation of the gait cycle adds further stress to an already vulnerable structure, potentially surpassing its injury threshold and leading to inflammation, pain, and tenderness more commonly at the medial tubercle of the heel bone and occasionally throughout the fascia. Furthermore, activities like prolonged jogging, running, or standing also increase the risk, although the precise aetiology of the condition remains unidentified [1, 5, 6]. Typically, the diagnosis of plantar fasciitis is established through a comprehensive history and clinical examination [5, 7]. The primary focus in managing plantar fasciitis primarily centres around pain alleviation. However, a significant concern for individuals coping with this condition is the functional limitation of the foot [8, 9]. Various therapeutic approaches have been employed to treat plantar fasciitis. Non-operative interventions, including medications such as Non-Steroidal Anti-inflammatory Drugs, corticosteroids, and platelet-rich plasma injections, are viable options [7, 8]. Physical therapy encompasses a range of modalities, such as moist heat packs, cryotherapy, and contrast baths for superficial treatments; therapeutic ultrasound, phonophoresis, iontophoresis, faradic foot baths, and advanced modalities like extracorporeal shock wave therapy, Light Amplification by Stimulated Emission of Radiation (LASER), light therapy, and matrix rhythm therapy for more profound treatments [10–13].

Additional therapeutic options include manual techniques, such as myofascial release and positional ease. Exercise therapy involves stretching, which helps to reduce the overactivity of the plantar fascia and the muscles of the posterior leg, while strength-building exercises help increase strength in the leg and small foot muscles [10, 11, 14]. If conservative treatments prove ineffective, more aggressive options like endoscopic or open plantar fasciotomy may be considered [14, 15]. In recent years, Class IV LASER has emerged as an effective therapeutic modality for the management of plantar fasciitis [2, 7, 16]. This non-invasive treatment method employs high power, 500 milliwatts or more [17]. It differs from low-level laser therapy (Class IIIb), which operates at lower intensities (< 500 mW) and is suited for superficial conditions. It utilizes higher-intensity LASER radiation to gradually and fractionally stimulate the absorption of energy by melanin and chromophores. The elevated power output is particularly crucial for addressing injuries in the deepest tissue layers, including fascia, muscles, tendons, ligaments, and cartilage. The increased energy emitted by class IV LASER is directed toward the tissues, where its optical energy induces dynamic vibrations [18, 19]. LASER emission exerts a positive influence on damaged tissue by promoting the proliferation of endothelial cells, enhancing angiogenesis, collagen synthesis, and fibroblastic activity. Furthermore, it suppresses anti-inflammatory cytokines, resulting in decreased inflammation. Additionally, LASER emission indirectly relieves pain by inducing the release of endogenous opioid peptides, which function as analgesics by blocking pain signals transmitted by A-delta and C-fibre nerves [2, 20–22].

Tape applied to the foot has been recommended to address biomechanical faults and to reduce the tension of plantar fascia, leading to alleviating symptoms related to plantar fasciitis. Low-dye taping is the most commonly used due to its functional purposes, despite differences in terminology [23, 24]. Excessive strain on the plantar fascia and repeated weight bearing can lead to tissue microtears, inflammation, and collagen degeneration, contributing to the pathology of plantar fasciitis [25]. The application of kinesio-taping can reduce fascial strain and could potentially promote tissue repair and decrease injuries for weight-bearing activities, forming the primary mechanical basis for taping treatment [26, 27].

Plantar fasciitis involves the plantar fascia contracting due to collagen structure changes and overuse, resulting in increased fascia activity and reduced extensibility. So, stretching of the plantar fascia helps to increase its flexibility and reduces the tone and hyperactivity of the fascia, hence limiting the windlass mechanism [28, 29]. There is a lack of evidence regarding the best stretching protocol, continuous or intermittent, to effectively reduce symptoms of plantar fasciitis. Moreover, in PF management, LASER therapy is not a gold-standard treatment but serves as an effective adjunct to conventional physiotherapy. The present trial addresses this gap by directly comparing Class IV LASER combined with two commonly used adjuncts, kinesio-taping and plantar fascia stretching. Therefore, this research aimed to compare the effectiveness of Class IV LASER therapy when given in combination with either continuous stretching, aided by low-dye kinesio-taping, or intermittent stretching, using a plantar fascia-specific stretching method, on the symptomatic presentation of

plantar fasciitis. The hypothesis was that the treatment might or might not significantly affect pain, pain pressure threshold, and functional disability.

Materials and methods

Study design and setting

This was a two-arm parallel group double-blinded Randomized clinical trial conducted at the outpatient department of Maharishi Markandeshwar Institute of Physiotherapy and Rehabilitation, MMDU, Mullana-Ambala, Haryana, India, between May 2023 to March 2024. The trial was registered with the Clinical Trial Registry of India (CTRI/2023/05/053275) and approved by the Institutional Ethics Committee (MMDU/IEC-2404). The research adhered to the principles of the Declaration of Helsinki, 2013, and the National Ethical Guidelines for Biomedical Research, 2017. The trial was conducted and reported in accordance with the CONSORT 2010 guidelines [30]. A completed CONSORT checklist is provided as supplementary material.

Participants

Participants of both genders, aged 18 to 45 years, who were diagnosed with plantar fasciitis, and reported plantar heel pain lasting at least more than 4 weeks and tenderness at the origin site of the plantar fascia on the calcaneum, with the intensity of the pain > 5 cm measured by visual analogue scale, were eligible for inclusion. Participants above 45 years were not included to reduce potential confounding from age-related degenerative changes in fascia, even though the condition's peak incidence extends up to 60 years. Participants with any other heel pathology (calcaneal stress fracture or plantar fascia neoplasm), any metal implant in the leg or foot, peripheral vascular diseases, neurological conditions, degenerative disorders, or skin disorders, being evaluated through a series of physical, radiological, and investigatory assessments by the physiotherapist, were excluded. Participants who had received any form of physiotherapy for plantar fasciitis within the last three months before enrolment were also excluded [31, 32].

Sample size

The sample size was estimated utilizing the statistical software G* Power 3.1.9.4 (Heinrich-Heine-Universität Düsseldorf, Germany) [33]. Based on the study by Akkurt F et al., a significance level of 0.05, a power of 90% (B = 0.9), and a reported mean between-group effect size across the three VAS measures of 0.9 were established [2]. The calculated sample size was 58 (including provision for dropouts; i.e., 29 participants per group), and participants were subsequently assigned to one of two groups through a random allocation process.

Randomization and blinding

The patients were randomly assigned to two groups that were, experimental group (class IV LASER therapy along with low-dye kinesio-taping) and the control group (class IV LASER therapy along with plantar fascia stretching) using a computer-generated table of randomized numbers. The participants remained blinded to the assigned intervention, which was ensured with concealed allocation, and the outcomes assessor was also blinded.

Procedure

Patients in the outpatient department with a diagnosis of plantar fasciitis were identified as potential participants. They underwent a comprehensive clinical evaluation by the concerned physiotherapist before the assessment of outcomes and commencement of treatment. This evaluation encompassed a review of the history, focusing on localized pain and tenderness over the medial tubercle of the calcaneus, along with the presence of sharp pain during the initial steps in the morning and after extended periods of inactivity. Specialized tests, including the windlass test (Specificity of 100% in differentiating plantar fasciitis) [34], along with the navicular drop test and the Silverskiöld test to rule out pathologies like flat feet and calf tightness, were also conducted to confirm the diagnosis [18]. Following affirmation of the diagnosis, they were presented with the proposal of the study and were assessed per the eligibility criteria. On fulfilling the criteria and agreeing to participate, written informed consent was obtained. Following that, each participant's demographic information and the outcome measures were assessed. Participants were then randomly assigned to one of the two therapy groups after completing the baseline evaluation. Following their two weeks of prescribed treatment, the final day of the treatment session saw the completion of the post-intervention evaluation for the outcome measures. To ensure participant adherence to the prescribed intervention and minimize participant dropout, regular monitoring of treatment sessions, maintaining

a follow-up diary, and telephonic reminders about the treatment sessions are required. After that, the obtained result was examined.

Outcome measures

The primary outcome measure was pain intensity, assessed using the visual analogue scale (VAS) and pain pressure threshold (PPT), measured with a digital pressure algometer. Secondary outcome measures included disability, measured with the Foot Function Index (FFI).

Visual analogue scale

The VAS is a subjective instrument, comprising a 10 cm long straight line, employed for the assessment of pain intensity. It has exhibited strong test-retest reliability, with an intraclass correlation coefficient (ICC) of 0.97 [35]. Each participant was asked to mark a point on a 10 cm long line that represented their pain level at its worst.

Pain pressure threshold

An algometer (ORCHID SCIENTIFIC) is a tool employed to establish the minimum pressure needed to induce pain by applying controlled pressure to a particular point on the body. It has a moderate to strong ICC value ranging from 0.752 to 0.874 [16, 17]. The pressure pain threshold was measured in Newton (N) at the midpoint of the anatomical site within the enthesis zone of the plantar fascia toward the heel in a standardized prone position, with their feet hanging from the examination table. Following a standardized site and procedures specific to the plantar fascia region minimizes the potential influence from myofascial trigger points. It was measured three times at each assessment, and the mean of these three readings was used for analysis.

Disability

The FFI measuring foot disability is a self-reported scale crafted to evaluate different facets of foot function from the patient's viewpoint. Its purpose is to gauge the influence of pain, disability, and activity limitations in individuals with foot problems. It consists of 17 items categorized into three subgroups. Patients rate each question on a scale from 0 to 10, where 0 signifies the absence of pain or dysfunction, and 10 represents the utmost level of pain or dysfunction. The minimal clinically significant difference for the FFI is established at 7 points, signifying a substantial alteration in foot function [36]. Additionally, the internal consistency of the FFI scores falls within a range of 0.73 to 0.96, indicating a high degree of consistency in the measurements [37]. Each participant was asked to self-report their function using the FFI, depending on their current ability to carry out the activities.

Interventions

The interventions included the employment of class IV LASER therapy to all the participants in both groups, along with the low-dye kinesio-taping technique in the experimental group or plantar fascia stretching in the control group.

Class IV LASER therapy

Participants were positioned in a prone posture with a pillow underneath the knees, and their feet were allowed to hang off the surface edge. Both the subject and the therapist wore eye protection to shield their eyes from rays. Subsequently, the application of the class IV LASER device (Lite Cure LCT1000 10-watt Cold Laser System) was executed using a handpiece positioned at a right angle to the treatment area. The power was set at 8 watts, energy at 8 J/cm^2, and a continuous wavelength frequency was applied **(**Table 1**)**. The convex head of the handpiece was moved in a continuous circular or linear motion, maintaining contact with the skin, starting from the heel and moving towards the plantar fascia direction. The speed of motion was 1 cm per second, with gentle pressure applied. These sessions were scheduled every other day over 2 weeks.

Low-dye Kinesio-taping technique

In the Experimental group, participants underwent a kinesio-tape application utilizing a low-dye taping technique on the plantar area of the foot. The patient was positioned in a

Table 1 Class IV LASER therapy parameters

Parameter	Specification
Device	Lite Cure LCT1000 (10-watt Cold Laser System)
Power Output	8 W
Energy Density	8 J/cm²
Wavelength	Continuous (Class IV, 980 nm)
LASER Medium	Gallium–Aluminum–Arsenide (GaAlAs) diode
Application Angle	90° to treatment area
Movement Pattern	Circular or linear motion
Movement Speed	1 cm/second
Application Area	From heel to plantar fascia
Pressure Applied	Gentle contact
Treatment Frequency	Every other day
Treatment Duration	2 weeks

relaxed supine position with the hip, knee, and ankle joints aligned in a neutral position, and the second toe in line with the other joints. A water-resistant, absorbent, and adhesive kinesiology tape was then employed. The target sites for taping were marked from the metatarsal joints to the posterior edge of the calcaneus bone. The tape was divided into two bands: the first band extended from the fifth metatarsal to the back side of the heel, and the second band extended from the first metatarsal to the back of the heel and a 50% stretch was applied to the middle portion of the tape, and the ends were applied without any stretch. The remaining four pieces of tape were applied with a 50% stretch in the middle third of the tape while the ends were left tension-free, running from the lateral to medial side in a dorsiflexed position of the foot. The foot was then placed in a neutral position, and the seventh tape was applied dorsally from the lateral to medial side with a 50% stretch in the middle third of the tape **(**Fig. 1**)**. This session was administered three days a week for 2 weeks [23].

Plantar fascia stretching

In the Control group, participants underwent plantar fascia stretching exercises. The stretching was performed in a seated position, with the affected leg crossed over the unaffected leg. The patient's hand was placed across the base of the toes, and the toes were gently pulled towards the shin. This position was held, followed by a return to the starting position. Five repetitions were performed with a 20-second hold. This exercise was repeated twice daily by the participants, and sessions were administered every day for 2 weeks [38].



Fig. 1 Low-Dye Kinesio-Taping Technique

Statistical analysis

Statistical analyses were conducted using SPSS for Windows, version 26.0. Categorical variables were analysed using descriptive statistics and frequency analysis, presenting counts. Numerical variables were summarized with mean and standard deviation. The normality of each demographic characteristic and outcomes for both groups was assessed using the Kolmogorov–Smirnov test of normality to ensure that the data followed a normal distribution, thereby meeting the assumptions required for parametric testing. As all the outcome measures and a few of the demographic characteristics followed a normal distribution, to evaluate within-group and between-group differences, parametric tests were employed: the paired t-test for within-group comparisons and the independent t-test for between-group comparisons. For the between-group analysis of the rest non-normally distributed demographic data, the Kruskal-Wallis test was utilized. The chi-square test was conducted to assess the baseline distribution of categorical data between the groups. The level of significance was set at $p \leq 0.05$. The minimally clinically important difference (MCID) value for pressure algometry in plantar fasciitis pain pressure threshold was calculated using the effect size × $SD_{Baseline}$ [39]. No intention-to-treat analysis was performed; dropouts were excluded from the final analysis.

Results

A total of 58 patients were enrolled in the study, with 29 subjects initially assigned to each group. However, 8 patients were lost to follow-up, resulting in a final sample size of 50 subjects. The results of these 50 subjects were analyzed, with 25 subjects in each group. Figure 2 illustrates the CONSORT flow diagram depicting the study profile.

Adherence to the study protocol

Four subjects in each group were lost to follow-up due to personal reasons. However, the remaining 25 subjects in both groups completed the scheduled exercise sessions for two weeks and participated in post-intervention assessments. As a result, complete data were available for all outcomes. The Class IV LASER therapy, low-dye kinesio-taping,

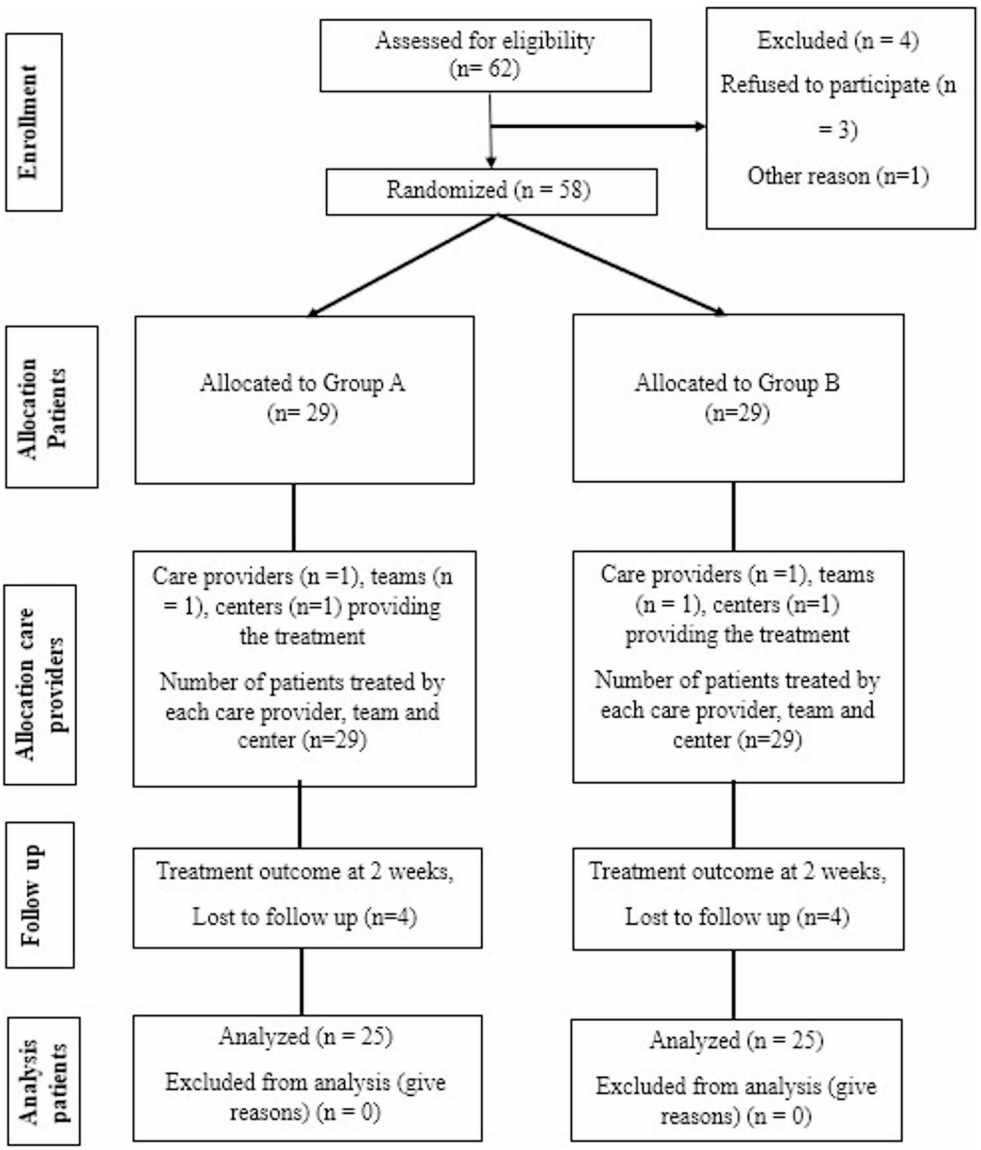


Fig. 2 CONSORT Flow Diagram

and plantar fascia stretching were found to be feasible and enjoyable for the subjects, and no adverse events such as skin irritation, redness, or burn with Class IV laser therapy and localized itching, allergic reactions, or blisters with low-dye kinesio-taping were reported during the sessions.

The baseline and demographic details for both groups are presented in Table 2. The demographic characteristics of the subjects were not normally distributed, except for weight and BMI. However, all outcome measures were normally distributed. Before treatment, at baseline, there were no statistically significant differences ($p > 0.05$) between the groups in terms of demographic characteristics and baseline dependent variables, suggesting homogeneity of data between the groups pre-intervention.

Table 2 Demographic and pre-intervention data

Variables	Experimental Group	Control Group	*p*-value
Age (Years)	30.20 ± 7.65	29.24 ± 7.41	0.654*
Height (cm)	162.68 ± 7.99	163.76 ± 7.44	0.623*
Weight (Kg)	62.16 ± 11.57	63.48 ± 8.87	0.653*
Gender (Male: Female)	13:37	13:37	0.753*
BMI (Kg/m^2)	23.47 ± 3.75	23.49 ± 2.62	0.983*
Side affected (Left: Right)	13:37	13:37	0.753*
VAS Pre (cm)	8.07 ± 0.65	8.18 ± 0.53	0.497*
PPT Pre (N)	18.52 ± 1.62	18.24 ± 2.06	0.592*
FFI Pre	57.83 ± 3.16	56.91 ± 2.62	0.270*

BMI, Body Mass Index; VAS, Visual Analogue Scale; PPT, Pain Pressure Threshold, Foot Function Index; Quantitative variables are shown as mean ± SD as the number. P-value showing the significance level which is set at > 0.05 (*).

Within group

Table 3 presents within-group comparisons of VAS, PPT, and FFI. Following 6 sessions of intervention, statistically significant differences ($p<0.05$) were observed in pain intensity (VAS) and PPT for both groups. It also depicts the within-group analysis of FFI, which reveals significant differences (p-value < 0.05) in both groups.

Between group

Table 4 and Fig. 3 demonstrate statistically significant differences (p-value < 0.05) between groups for VAS, PPT, and FFI. The observed changes were significantly more favourable for the Control group, with a greater reduction in pain intensity, a significant increase in pain pressure threshold, and a more pronounced decrease in foot pain-related disability.

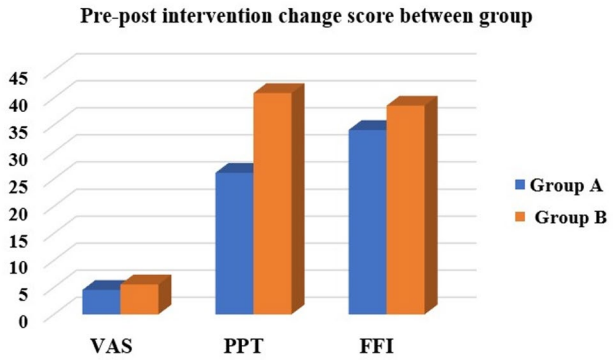


Fig. 3 Change score of VAS, PPT and FFI post-pre difference values between the groups

Discussion

To the authors' knowledge, this is the first randomized clinical trial that investigated the effect of combining class IV LASER therapy along with a kinesio-taping or plantar fascia stretching program of 2 weeks on pain intensity, PPT, and disability in individuals diagnosed with plantar fasciitis. Between-group differences in primary outcomes (Cohen's d 1.2–2.4) and secondary outcomes (Cohen's d 0.7) were observed. Both groups improved, but the control group exhibited more significant improvement than the experimental group for all outcome measures. The study's main finding suggests that both treatments are efficient in treating plantar fasciitis. However, for more pronounced symptom reduction in a shorter duration and with minimal or no side effects, class IV LASER therapy and plantar fascia stretching have been clinically proven to be more successful in ameliorating plantar fasciitis subject's symptoms.

The synergistic interaction between class IV LASER therapy and the adjunctive interventions applied in this study may partly explain the superior outcomes observed. Photobiomodulation from Class IV LASER enhances cellular metabolism, microcirculation, and mitochondrial ATP synthesis, thereby accelerating tissue repair and modulating inflammatory mediators [40]. When used in combination with Kinesio-taping, these physiological effects may potentiate the tape's mechanical lifting action on the skin, improving lymphatic drainage, reducing local edema, and facilitating pain relief [41]. The improved microcirculation and soft-tissue extensibility achieved through LASER therapy can further enhance the proprioceptive and unloading effects of taping on the plantar fascia.

Similarly, the concurrent use of plantar fascia stretching with Class IV LASER therapy may amplify the effects of stretching by increasing fascial temperature and elasticity, promoting collagen realignment, and

Table 3 Within-group comparison

Experimental Group				Control Group		
Variables	Baseline	At 2 weeks	p-value	Baseline	At 2 weeks	p-value
VAS (cm)	8.07 ± 0.65 18.52 ± 1.62	3.54 ± 0.65 44.67 ± 6.49 23.75 ± 8.32	0.001*	8.18 ± 0.53	2.63 ± 0.46	0.001*
PPT (N)	57.83 ± 3.16		0.001*	18.24 ± 2.06	59.15 ± 5.28	0.001*
FFI			0.001*	56.91 ± 2.62	18.32 ± 2.51	0.001*

VAS, Visual Analogue Scale; PPT, Pain Pressure Threshold; FFI, Foot Function Index; Quantitative variables are shown as mean ± SD; Level of significance (*) set at p-value < 0.05.

Table 4 Between-group comparison

Post - Pre intervention comparison						
Variables	Experimental Group	Control Group	Mean difference	t-value	*p*-value	Effect size
VAS (cm)	4.53 ± 0.97	5.55 ± 0.63	1.02	4.40	0.001*	1.2
PPT (N)	26.15 ± 6.03	40.91 ± 5.84	14.76	8.78	0.001*	2.4
FFI	34.07 ± 8.57	38.58 ± 3.42	4.51	2.44	0.018*	0.7

VAS, Visual Analogue Scale; PPT, Pain Pressure Threshold; FFI, Foot Function Index; Quantitative variables are shown as mean ± SD, Mean difference and t-value; level of significance set at p-value < 0.05.

reducing stiffness [38]. The photothermal and photochemical actions of LASER irradiation enhance tissue pliability, thereby allowing greater stretch tolerance and functional improvement.

Stretching effects

The improved outcomes in the Control group can be attributed to the active, physiological responses triggered by stretching exercises. Plantar fascia stretching involves intermittent, active stretches that produce neuromuscular and biochemical changes, offering benefits beyond the passive support of taping. When the plantar fascia is stretched, sensory receptors, such as muscle spindles, detect variations in muscle length and the speed of the stretch. These receptors send signals to the spinal cord and brain, activating the stretch reflex to stabilize muscle length and prevent overstretching injuries. This neuromuscular response not only preserves muscle tone but also promotes dynamic structural adaptations over time [42].

Plantar fascia stretching is crucial for elongating and remodelling the contracted fascial tissue, thus restoring proper foot mechanics. By replicating the windlass mechanism, which involves tightening the plantar fascia during toe dorsiflexion, stretching exercises improve fascial flexibility and help redistribute mechanical loads across the foot. As a result, tissue strain decreases, microtrauma is reduced, and weight-bearing function improves [28, 29]. Systematic reviews and clinical studies have consistently shown that plantar fascia stretching results in significant improvements in both pain relief and functional performance, often matching or surpassing the benefits of other conservative treatments [7, 43]. The repetitive and dynamic nature of stretching helps reinforce biomechanical corrections and promotes long-term adaptations of both the fascial and muscular structures. Stretching not only addresses the mechanical issues in plantar fasciitis but also activates various neuromuscular and circulatory mechanisms that accelerate tissue recovery, reduce pain more efficiently, and enhance overall foot function at a faster rate than passive taping methods.

These improvements may also be related to the enhanced local circulation and the removal of substances contributing to hyperirritability and pain [7]. Previous literature demonstrated that combining manual therapy with self-stretching for plantar heel pain is more effective than self-stretching alone [44]. The current study supports these findings, as stretching can improve pain by preserving the length-tension relationship of fibres. In the control group, plantar fascia stretching may have improved disability by altering the length-tension relationship of the contracted fascia, allowing it to return to its normal activation function [29]. Stretching of the myofascial structures is postulated to trigger the release of actin and myosin heads, leading to an accumulation of adenosine triphosphate and improved vasomotor response and circulation. This helps remove substances that contribute to pain, reducing pain levels. Stretching techniques can thus effectively improve vasomotor response, blood flow, nociceptive stimulus, and proprioception in patients with plantar fasciitis [45].

Low-dye Kinesio taping

On the other hand, the low-dye Kinesio taping used in the Experimental group provides a continuous, passive external support. It works by offloading the plantar fascia and addressing abnormal foot mechanics, especially excessive pronation. While it provides short-term stability and can deliver immediate pain relief, the benefits of taping are largely dependent on its duration of application. The mechanical unloading effect ends once the tape is removed, making the benefits temporary unless reapplied consistently over a longer period. Additionally, Kinesio taping does not directly promote intrinsic muscle or fascial adaptations. It relies on external mechanical support and proprioceptive feedback, which may not be sufficient for the long-term resolution of plantar fasciitis unless paired with active treatments. Research suggests that extended taping up to six weeks may be necessary to support neuromuscular re-education and foster lasting biomechanical improvements. However, even then, it lacks the direct stimulation of tissue remodelling and vascular responses that active stretching provides [46].

Previous studies have shown that both low-dye taping and fascia taping can relieve plantar heel pain in runners by relaxing the contracted fascia. Kinesio-taping has also been demonstrated to reduce pain by other studies [47, 48]. Taping helps alleviate mechanical stress on the plantar fascia. Low-dye kinesio-taping, applied from the origin to the insertion point of the plantar fascia with the foot dorsiflexed, can prevent microtrauma and promote normal foot mechanics. A study found that combining low-dye kinesio-taping with extracorporeal shockwave therapy (ESWT) improved foot function compared to sham taping with ESWT [49].

A study by Park C et al. found that kinesio-taping enhances foot stability and accelerates recovery. However, the effects of taping are only sustained while the tape is applied. Long-term application is necessary for neuromuscular re-education and avoiding abnormal biomechanical forces. Literature indicates that prolonged kinesio-taping for 6 weeks can only enhance foot function by guiding the fascia in a desired direction of movement [46]. Given the

absence of a long-duration kinesio-taping intervention protocol, a short-term stretching protocol was possibly found to be more beneficial for treating plantar fasciitis.

Class IV LASER

Class IV LASER therapy, on the other hand, improves local circulation and triggers photochemical changes, leading to pain relief. Previous research has shown that Class IV LASER therapy combined with silicone insoles is more effective for treating plantar fasciitis than silicone insoles alone, likely due to deeper penetration and enhanced healing [2]. The current study found that kinesio-taping and Class IV LASER therapy reduced pain in the experimental group, thus aligning with previous research. This therapy also improved disability by reducing cellular stress and promoting fascia healing. Literature supports Class IV LASER's effectiveness in enhancing foot function through collagen synthesis and tissue repair [2, 7, 16]. However, a study by Tckoz et al. found no significant improvement in foot function with Class IV LASER therapy in patients with plantar fasciitis and heel spurs [50]. This may be due to the heterogeneity of the patient population. Another study by Naruseviciute et al. compared Class IV LASER to low-intensity LASER therapy for unilateral plantar fasciitis, finding Class IV LASER to be more effective in reducing plantar fascia thickness but not pain [27]. As compared to low-level LASER therapy, possibly the enhanced depth of penetration and greater energy density achieved with Class IV LASER may yield more pronounced reductions in pain and improvements in soft-tissue flexibility in PF.

Clinical significance

The MCID scores for the experimental group and the control group were 7.2 N and 18.2 N, respectively, indicating that a change of at least 7.2 N in PPT for the experimental group and 18.2 N for the control group is considered a clinically significant improvement. The average PPT difference was 26.1 N in the experimental group and 40.9 N in the control group, both demonstrating substantial clinical improvement. The mechanism behind the promising improvement by adhering to a short-term protocol for plantar fasciitis subjects in the experimental group could be attributed to a decrease in the pulling force on the plantar fascia and enhancement in local circulation, thereby potentially preventing repetitive microtrauma. Additionally, direct stimulation of nociceptors and mechanoreceptors might have also contributed to pain relief and improved PPT [51]. Clinically, the MCID of VAS in plantar fasciitis is generally reported as 9 mm on a 100-mm visual analogue scale, and for FFI, a change of about 6.5 points is taken to be clinically significant [52]. In this study, both groups surpassed these MCID values, with the control group having a more significant decline in VAS scores and greater improvement in FFI than the experimental group. Additionally, the ICC for the assessor in this trial was not calculated due to study design constraints.

Strengths and limitations

This study has strengths and limitations. Strengths include its focus on comparing plantar fascia stretching along with low-dye kinesio-taping or Class IV LASER therapy without incorporating conservative treatments, its evaluation of PPT as a primary outcome, and its potential to expand physiotherapists' understanding of adjunct treatment methods for plantar fasciitis.

Limitations include the lack of a pure control group, the absence of tissue-level evaluation using diagnostic ultrasound or biomarkers, and the short duration and limited sample size due to patient availability, which precluded long-term follow-up. Moreover, a subgroup analysis by gender for all outcomes was considered; however, due to the relatively small sample size and unequal group distribution, it was not statistically feasible to derive clinically meaningful conclusions, and thus couldn't be performed. Further, the pain intensity was assessed as participants' self-reported 'worst pain' without specifying the timing, which may have introduced variability and limited the precision of functional pain assessment.

Future recommendation

Clinical trials are needed to investigate the effectiveness of combining kinesio-taping and rigid taping with Class IV LASER therapy. The primary outcome could include the measurement of quality of life. Follow-up sessions may be included to assess the long-term impact of the intervention.

Conclusion

The study concluded that both kinesio-taping combined with the class IV LASER therapy group and plantar fascia stretching combined with the class IV LASER therapy group contributed to pain reduction, improved PPT, and enhanced foot function. However, the plantar fascia stretching group demonstrated higher treatment efficacy for all variables compared to the kinesio-taping group. Therefore, it is inferred that patients with plantar fasciitis may respond better clinically when Class IV LASER therapy is paired with plantar fascia stretching.

Author contributions Author's contribution:1.Sarita Yadav (Primary author), conducted the research, provided research materials, acquired, and analysed the data.2. Sunita Sharma (Corresponding author), contributed in concept and design of the study, data analysis, and interpretation of data; drafting the article or revising it critically for important intellectual content; Final approval of the version to be published.3. Shikhar Singh Thakur, Acquisition of the data, critical revision of the manuscript, final approval of the version to be published; and Aptitude to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.4. Sandeep Pattnaik, Drafting the article or revising it critically for important intellectual content, Final approval of the version to be published.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval The research was approved by the Institutional Ethics Committee (IEC-2404) and carried out in compliance with the principles of the Declaration of Helsinki (2013) and the National Ethical Guidelines for Biomedical Research (2017).

Informed consent Informed consent was obtained from all participants prior to their involvement in the study, ensuring they fully understood the purpose, procedures, potential risks, and their right to withdraw at any time.

Competing interests The authors declare no competing interests.

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